2.4 Modeling with Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.4 · Student sheet

I can…

- write quadratic equations using vertex form and intercept form.
- use finite differences to tell whether data are linear, quadratic, or neither.
- model real-life data and situations with quadratic functions.

Vocabulary · fill in the missing word

Vertex form — $y = a(x - h)^2 + k$: use when you _____ the vertex and one more point

Intercept form — $y = a(x - p)(x - q)$: use when you know the _____ and one more point

First differences — the differences between _____ y-values (equally spaced x)

Second differences — the differences of the first differences; constant second differences mean _____ data

Part 1 · Ball toss data

Time (s)	0	0.5	1	1.5	2	2.5
Height (ft)	6	22	30	30	22	6
First differences						
Second differences						

Type of function: _____ When is the ball highest? _____ About how high? _____

Part 2 · Notes: fill in the blanks

Know the vertex: $y =$ _____. Know the x-intercepts: $y =$ _____.

Constant first differences → _____; constant second differences → _____.

Part 3 · Examples (work them with the class)

Example 1

Write an equation of the parabola with vertex $(2, 5)$ that passes through $(4, -3)$.

Example 2

Write an equation of the parabola with x-intercepts -1 and 4 that passes through $(2, 12)$.

Example 3

Tell whether the data are linear, quadratic, or neither. Write a function for the data.

x	0	1	2	3	4	5
y	2	5	12	23	38	57

Example 4

You have 80 feet of fencing to make a rectangular pen against a barn (the barn is one side). What dimensions give the greatest area? What is that area?

Part 4 · Partner practice

Partner 1

Write an equation of the parabola with vertex (−3, 1) that passes through (−1, 9).

Partner 2

Write an equation of the parabola with x-intercepts 0 and 6 that passes through (3, 18).